Bridging Domain Gaps in Visual Correspondence

From Multi-Modal Fusion to Data-Curated Motion Latents

Spencer Low · PhD Dissertation Proposal · BYU Computer Science · April 2026

Chapter 1: Cross-Modal Aerial Sensing

Chapter 2: Directed Coverage Metrics

Chapter 3: Coverage-Based Data Curation

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Visual Correspondence: One Problem, Many Domains

Dense Correspondence

Given a point in image A, find its counterpart in image B. Underlies: optical flow · semantic matching · 3D reconstruction · tracking · navigation

Optical Flow

Semantic Match

3D Reconstruct

Cross-Modal

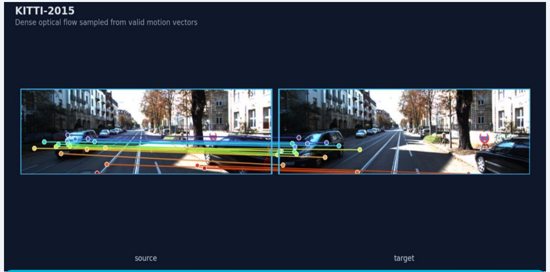
The Core Challenge: Domain Shift

- Training data ≠ deployment data
- Sensing modality gaps (EO → SAR → IR)
- Motion statistic mismatches
- Supervision format heterogeneity
- Dense flow vs. sparse keypoints

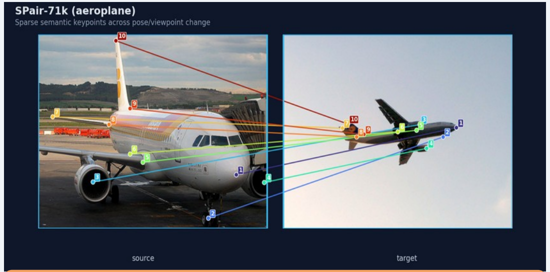
Train on	*Deploy to*
FlyingThings (synthetic)	**KITTI-2015 (real driving)**
EO (visible light)	**SAR (radar backscatter)**
PointOdyssey (tracking)	**SPair-71k (semantic)**
SPair-71k (semantic)	**PF-PASCAL (semantic)**

Correspondence Tasks Across Vision Domains

All three dissertation chapters address variants of the same fundamental challenge: find matching points across mismatched images.



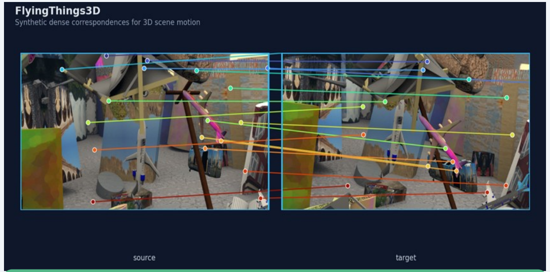
Optical Flow



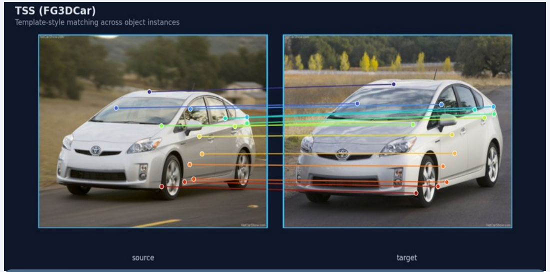
Semantic Matching

Dense Tracking

Cross-Modal



3D Scene Flow



Template Matching

Ch. 1: cross-modal (SAR ↔ EO) · Ch. 2: metric & synthetic data design · Ch. 3: training data curation for all task families

Dissertation Overview

Appearance alignment between training and deployment data is a weaker predictor of correspondence transfer than is commonly assumed. The geometric structure of motion — how displacements are distributed across directions, magnitudes, and scene locations — is the more reliable and actionable axis: it explains transfer variance that appearance metrics cannot, predicts zero-shot performance across diverse benchmarks, and provides principled selection signals from whole datasets down to individual pairs.

Ch. 1

Cross-Modal Aerial Sensing

EO ↔ SAR features transfer despite extreme visual disparities — appearance-domain match is not required. Progress is bottlenecked by training data scale, motivating synthetic augmentation.
Published (CVPRW 2022–2024)

Ch. 2

Directed Coverage Metrics

Non-photorealistic fractals match photorealistic baselines zero-shot — motion geometry, not appearance, is the key predictor. Directed BFV coverage separates failure modes that symmetric metrics conflate.
Under review (ECCV 2026)

Ch. 3

Coverage-Based Data Curation

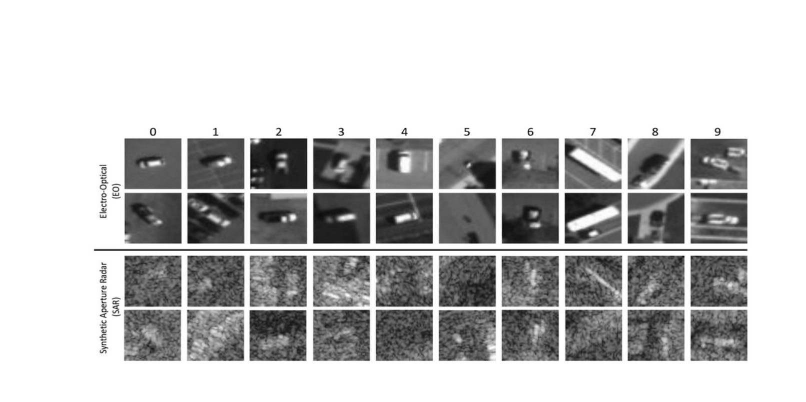
BFV motion coverage as utility function curates heterogeneous training pools — outperforms random sampling, recovers expert source routing without source labels.
In progress

Appearance alignment is increasingly saturated by pre-trained features. Motion geometry is the underutilized axis — and each chapter adds evidence.

Chapter 1: Cross-Modal Representation Learning for Aerial Imagery

Motivation: the most severe form of domain shift — different sensing physics

UNICORN: Co-registered EO + SAR chips (10 vehicle classes)



SAR→EO Translation (MAVIC-T)

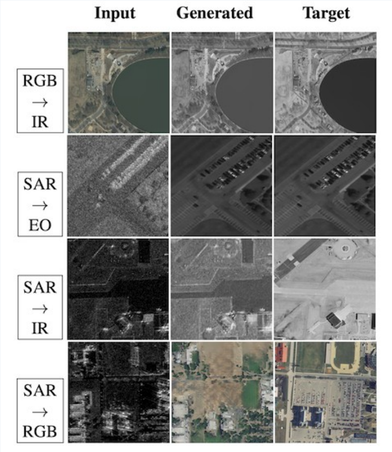
Key Contributions & Findings

- MAVOC/MAVIC-C: SAR+EO classification benchmark series (CVPRW 2022–2024), geographic held-out splits exposed severe location overfitting
- MAVIC-T: Multi-directional translation (SAR→EO, SAR→RGB, SAR→IR, RGB→IR); SAR→RGB substantially harder than SAR→EO
- MAGIC dataset: Aligned SAR · RGB · IR stacks for cross-modal training & evaluation
- MINNIMAN: Magnetic anomaly map prediction from EO+hyperspectral via VQ-GAN; proof-of-concept for GPS-denied navigation

🔑 **Appearance similarity did not predict transfer as strongly as expected; cross-modal supervision and shared structure mattered more than looking visually closer.**

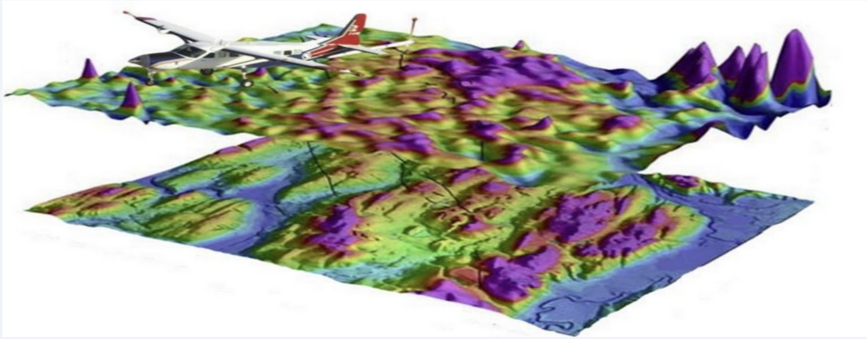
Chapter 1: MAVIC-T (Benchmark) + MINNIMAN (Cross-Domain Inference)

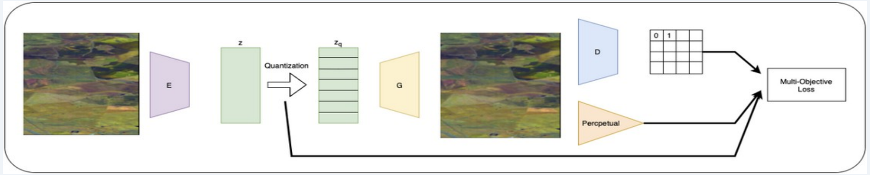
MAVIC-T — Open-Source Translation Benchmark



- 4 directions: SAR→EO, SAR→RGB, SAR→IR, RGB→IR · MAGIC dataset
- SAR→RGB substantially harder — larger sensing physics gap
- Metric: L1 + LPIPS + FID · Geographic disjoint splits exposed location overfitting

MINNIMAN — Magnetic Map Inference for GPS-Denied Nav





- Predicts magnetic anomaly maps from EO + hyperspectral inputs
- VQ-GAN codebook discretizes map space · teacher-student for modality changes
- Proof-of-concept: same cross-modal logic, applied to navigation signals

MAVIC-T = open benchmark (CVPRW 2022–2024) · MINNIMAN = our system · Both published & complete.

From Chapter 1 to Chapter 2: Why Labeled Data Is Not Enough

The Aerial Sensing Challenge

- Sensors capture at different resolutions, projection models, and ground footprints — spatial alignment is non-trivial
- Temporal co-registration requires the scene to be stable across passes — rare in practice
- MAGIC is one of the few open-source datasets spatially and temporally co-registered across SAR · RGB · IR · hyperspectral
- Despite this, scale and geographic diversity remain limited — and dense correspondence labels do not exist

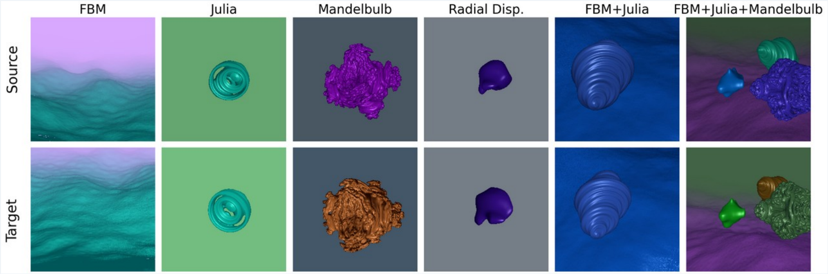
Correspondence Labels Are Expensive Everywhere

- Dense optical flow: requires calibrated stereo rigs or LiDAR depth — hardware-constrained
- Semantic keypoints: human annotators per pair — slow, expensive, domain-specific
- Labels for driving ≠ labels for aerial ≠ labels for medical — no single dataset generalizes
- The bottleneck is not modeling — it is labeled data

Chapter 2 Approach: generate unlimited labeled correspondence pairs synthetically — and ask whether controllable motion structure predicts real-world transfer

Chapter 2: SDF-Fractal3D — A Controllable Synthetic Pipeline

Intentionally non-photorealistic — yet transfers strongly zero-shot across diverse real benchmarks



Design Principles

- Asset-free, procedurally scalable — no 3D assets or rendering pipelines
- Factorized: motion sampler × appearance stage are independent
- Motion sampler: camera viewpoints + controllable zoom distribution
- Appearance: procedural textures, independent lighting & backgrounds
- Allows motion-only interventions while holding appearance fixed
- 2D Warp variant: same 3D correspondence field, warped photor

Existing synthetic datasets require:

FlyingThings3D

Pre-rendered offline
3D asset library download required
Photo-real textures & lighting

Kubric

Pre-rendered offline
Asset pipeline download required
Realistic physics, reflections & shadows

No real imagery, no 3D assets, no photorealistic textures. Does it transfer? Let's find out.

Zero-Shot Transfer Grid: SDF-Fractal3D Competitive Despite Non-Photorealism

PCK @ α=5%: a keypoint prediction is correct if within 5% of the image diagonal from ground-truth. RAFT + CATs++ averaged — zero-shot, no fine-tuning on any target.

Train Set	Type	KITTI-12	KITTI-15	SPair	TSS	PF-PASCAL	PF-WILLOW	PointOd.	FlyingThings	Middlebury	SDF-Fractal3D
Sintel	Flow	94.8	91.4	10.0	44.6	27.9	30.7	59.7	73.5	53.8	26.8
FlyingThings3D	Flow	**96.2**	93.4	10.0	44.8	**29.0**	31.6	55.0	**80.6**	53.9	27.0
SPair-71k	Semantic	40.4	41.9	**13.0**	27.6	18.1	21.7	23.6	28.5	23.3	17.7
PointOdyssey	Tracking	91.2	88.9	10.5	41.4	25.6	28.2	**70.4**	68.9	**54.1**	25.6
ImageNet 2D Warp	Flow	68.5	72.7	10.0	34.4	20.5	23.9	36.4	56.3	41.1	24.2
SDF-Fractal3D	Flow	89.0	90.5	11.3	42.7	27.4	29.9	46.6	68.9	48.1	34.0
SDF-Fractal3D (2D Warp)	Flow	86.1	88.9	11.6	39.3	27.7	30.0	52.0	70.8	49.2	29.8
SDF-Fractal3D (Zoom) ★	**Flow**	95.7	**94.6**	11.0	**46.3**	28.4	**32.0**	48.8	72.3	51.6	**54.1**

SDF-Fractal3D (Zoom): 94.6 KITTI-15 — matches FlyingThings without real imagery

SDF-Fractal3D (Zoom): 46.3 TSS — non-photorealism doesn't prevent motion transfer

SPair-only: strong semantic (PF-PASCAL) but weak optical flow — no single source dominates

FlyingThings wins some columns, SDF-Zoom wins others — realistic rendering alone is not the determining factor

🔑 **Fractals match photorealistic baselines on multiple benchmarks. Appearance is not the explanation — something else is driving transfer.**

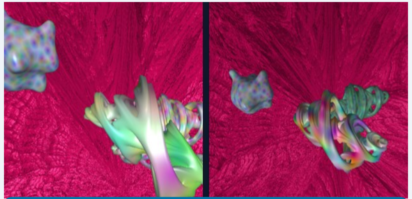
What the Training & Evaluation Datasets Look Like

SDF-Fractal3D has no real imagery — yet it just matched photorealistic baselines. Look at what the benchmarks actually look like.

Training Sources

SPair-71k

PointOdyssey



SDF-Fractal3D

FlyingThings3D

Evaluation Targets

KITTI-2015

PF-PASCAL

PF-WILLOW

TSS (3 subsets)

🔑 **Fractals look nothing like any of these — no roads, no cars, no animals. If appearance isn't the signal, what is?**

Representing Datasets: Bag-of-Flow-Vectors (BFV)

Motion Descriptor

$$f_{flow} = [\ x_{src},\ y_{src},\ \Delta x,\ \Delta y\ ]$$

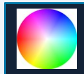
All components normalized by image dimensions · One 4-D sample per correspondence

- Resolution-agnostic — works across datasets with different image sizes
- Captures where motion occurs AND how objects move
- No rigid spatial grid → avoids discretization artifacts
- Physically meaningful Euclidean scale → enables ε-coverage radius

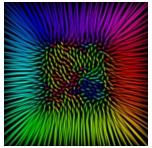
Appearance: patchwise DINOv2 embeddings projected onto unit hypersphere

Each dataset becomes a point cloud in BFV space — transferability becomes a geometric coverage problem.

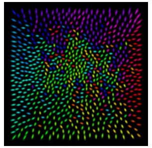
Representative BFV Splats



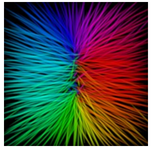
Training corpora



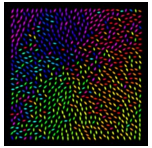
SDF-Fractal3D



PointOdyssey

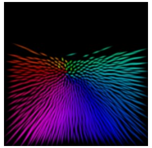


SPair

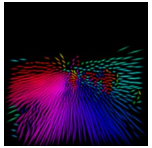


FlyingThings

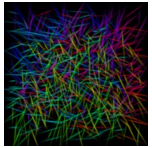
Benchmark / eval sets



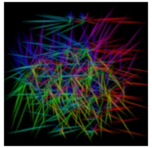
KITTI-2012



KITTI-2015



PF-PASCAL



PF-WILLOW

Directed Coverage: An Asymmetric Diagnostic Framework

ε-Support Inclusion: $S_\varepsilon(A \mid B) = \frac{1}{|A|}\sum_{a\in A} \mathbf{1}[\min_{b\in B}\|a-b\|_2 \le \varepsilon]$

HOF = Histogram of Optical Flow (angle bins)

4 predictors: BFV Sε(Eval | Train) · BFV Sε(Train | Eval) · DINOv2 Sε(Eval | Train) · DINOv2 Sε(Train | Eval)

Eval ⊆ε Train

Target-Support View

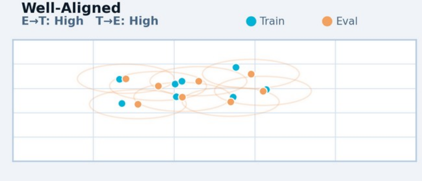
What fraction of eval BFV points have a training neighbor within ε?
Low = model never encounters those motion modes → transfer fails.

Train ⊆ε Eval

Off-Target-Mass View

What fraction of training BFV points have an eval neighbor within ε?
Low = training wastes capacity on off-target motion patterns.

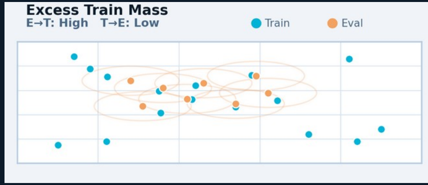
The 4 Coverage Regimes:



Eval ⊆ε Train: High · Train ⊆ε Eval: High

Well-Aligned

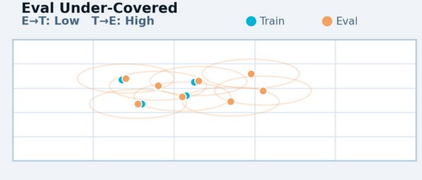
Training covers eval and is efficiently used. Best transfer outcome.



Eval ⊆ε Train: High · Train ⊆ε Eval: Low

Excess Train Mass

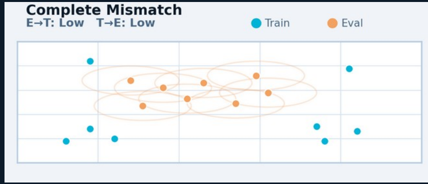
Eval is covered but training has many irrelevant motions. Wasteful but OK transfer.



Eval ⊆ε Train: Low · Train ⊆ε Eval: High

Eval Under-Covered

Eval requires motion modes absent from training. Transfer will fail.



Eval ⊆ε Train: Low · Train ⊆ε Eval: Low

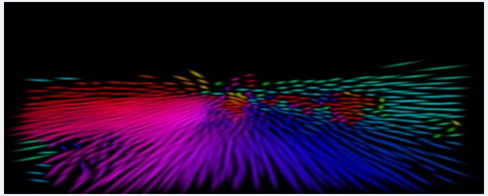
Complete Mismatch

Neither direction covered. Severe domain gap.

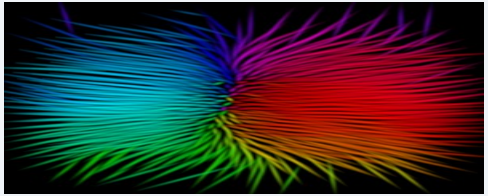
🔑 **Directed support inclusion separates failure modes that KL-div and MMD conflate. KL-div is also numerically unstable (requires density estimation); ε-**

Utility-Guided Motion Interventions: Coverage Predicts Transfer

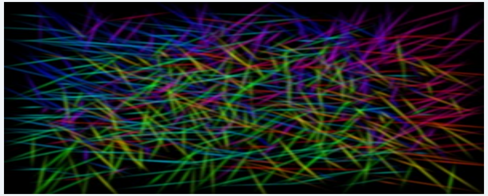
Appearance held fixed (‖ΔA‖ ≈ 0) — only the motion sampler varies. Each intervention is designed to target specific benchmark motion distributions.



KITTI-2015



TSS



PF-PASCAL

Base / Zoom / Flip — PCK @ α=5%

Training Variant	Type	PCK@5%	Δ PCK	F(E→T) coverage ↑
Eval = KITTI-2015				
SDF-Fractal3D	base	90.43	+0.00	0.197
+ zoom ★	**zoom**	**94.94**	**+4.51**	**0.653**
+ flip	flip	84.72	−5.71	0.132
Eval = TSS				
SDF-Fractal3D	base	46.58	+0.00	0.061
+ zoom ★	**zoom**	**53.42**	**+6.84**	**0.070**
+ flip	flip	39.89	−6.69	0.044
Eval = PF-PASCAL				
SDF-Fractal3D	base	30.72	+0.00	0.031
+ zoom ★	**zoom**	**34.57**	**+3.85**	**0.048**
+ flip	flip	22.27	−8.44	0.029

🔑 Motion is the causal lever: zoom into target motion space → PCK rises; flip mismatch → PCK drops sharply. Appearance is constant throughout — BFV coverage is an actionable intervention signal.

Transferability Estimator: Predicting Transfer Without Retraining

Ridge-regularized linear model · predicts within-context residual performance · evaluated under 3 leakage-safe protocols

0.70

Pairwise Ranking Accuracy (c-index)

vs. 0.50 chance baseline

E ⊆ε T

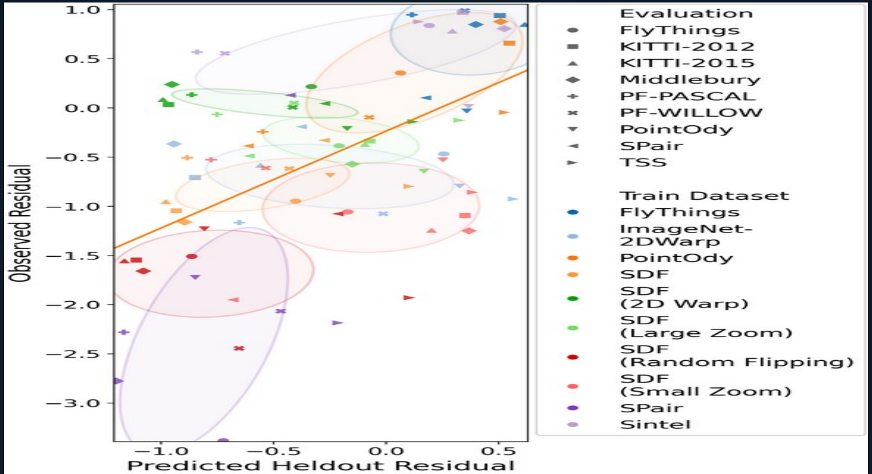
Most Consistent Predictor

Eval subset of training — best across all 3 held-out protocols

3

Strict Held-out Protocols

Train, Eval, Both



Configuration	Predictors	Hold-out Train	Hold-out Eval	Hold-out Both
Symmetric (baseline)				
Appearance MMD	MMD×1	46.2%	44.3%	41.3%
Flow MMD	MMD×1	58.7%	58.7%	58.7%
Directed – single direction				
Flow (Eval→Train)	F×1	61.7%	61.7%	61.7%
Appearance (Eval→Train)	A×1	54.0%	58.7%	54.6%
Directed – bidirectional				
Flow, bidirectional	F×2	65.4%	65.6%	65.5%
Appearance, bidirectional	A×2	63.5%	63.9%	61.9%
Best model				
Flow + Appearance bidir. ★	F×2+A×2	70.1%	69.8%	69.4%

Chapter 2 Summary & Bridge to Chapter 3

Contributions (ECCV 2026, under review)

- BFV motion descriptor: resolution-agnostic 4D representation in Euclidean space
- Directed coverage (Eval $\subseteq_{\varepsilon}$ Train, Train $\subseteq_{\varepsilon}$ Eval): separates missing support from off-target mass — symmetric MMD cannot
- SDF-Fractal3D: non-photorealistic synthetic pipeline; motion structure alone drives transfer
- Transferability estimator: 0.70 pairwise ranking accuracy across 3 strict held-out protocols
- Diagnostics support intervention — same signal drives Chapter 3 curation

Bridge to Chapter 3

Ch. 2 asks at the dataset level:
"Does dataset A transfer better than B to target T?"

> *Ch. 3 asks at the pair level:*
> *"Which specific training pairs from a mixed pool best support transfer to multiple targets simultaneously?"*

- ClusterCov is the first probe: does BFV coverage-based selection help at all?
- GIST is the full framework: BFV ε-coverage utility + diversity + approximation guarantee
- Eval $\subseteq_{\varepsilon}$ Train is the reward signal — measures missing target support

frontier

🔑 **The diagnostic from Ch. 2 becomes the objective function in Ch. 3.**

Chapter 3: Data Curation for Dense Correspondence

Which training pairs — from a single source or across many — best support transfer to target benchmarks?

Even within a single source

- PointOdyssey: 3 clips × ~1,800 frames → ~4M candidate pairs
- Consecutive frames share nearly identical flow statistics
- Exhaustive training is impractical — massive within-source redundancy
- ClusterCov is the first probe: does motion-space selection beat random?

Even harder across heterogeneous sources

- Pool: PointOdyssey (98.7%) · SPair-71k (1.2%) · PF-PASCAL (0.07%)
- Dense flow vs. sparse keypoints: incomparable supervision density
- Naive pooling over-rewards densest source by volume not usefulness
- No principled cross-source budget allocation without explicit criteria

Candidate pool composition:

PointOdyssey	SPair-71k	PF-PASCAL
98.7% | ~4.3M pairs	1.2% | ~52k pairs	0.07% | ~3k pairs

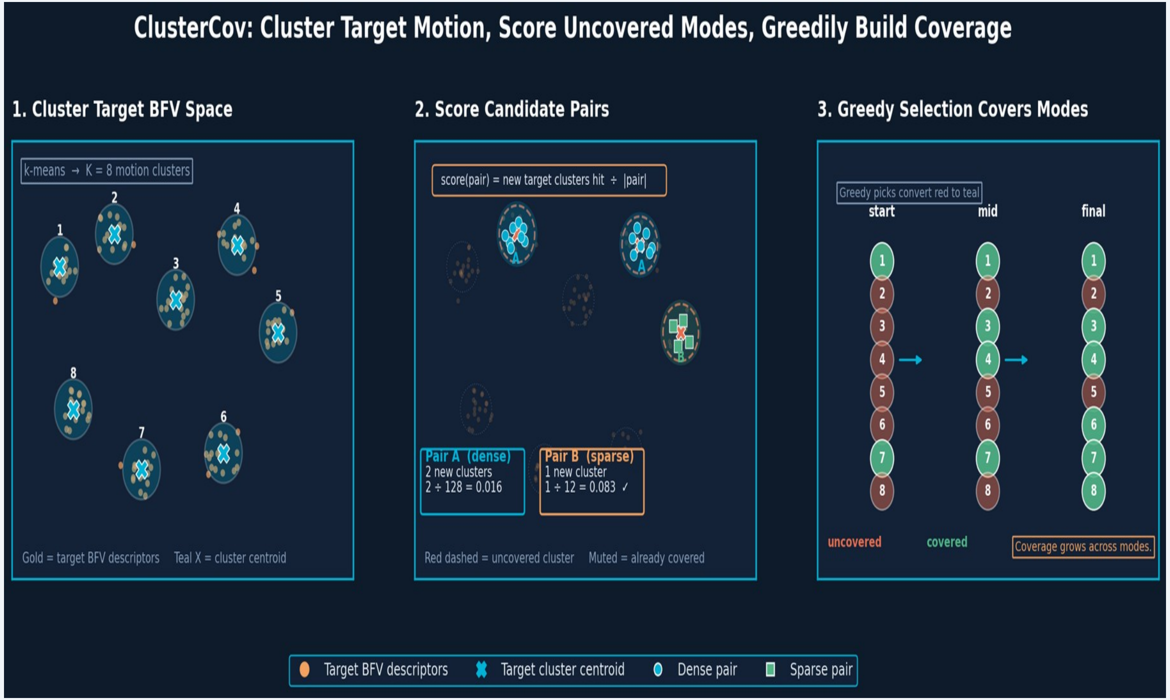
- ClusterCov establishes that coverage-based curation is worth doing
- GIST is needed to add guarantees, explicit diversity control, and principled heterogeneous allocation

ClusterCov: Preliminary Coverage Probe

Preliminary selector: test whether BFV coverage alone is already enough to beat random curation.

1. **Cluster target BFV space**
   K-means on target benchmark correspondences → motion cluster centroids
2. **Score candidate pairs**
   Each training pair scores by how many uncovered target clusters it reaches within radius ε
3. **Greedy selection**
   Iteratively select highest-scoring pair, mark covered clusters, repeat to budget
4. **Normalize by supervision density**
   Divide coverage gain by # valid correspondences in pair → pragmatic probe, not a formal guarantee

Visual intuition

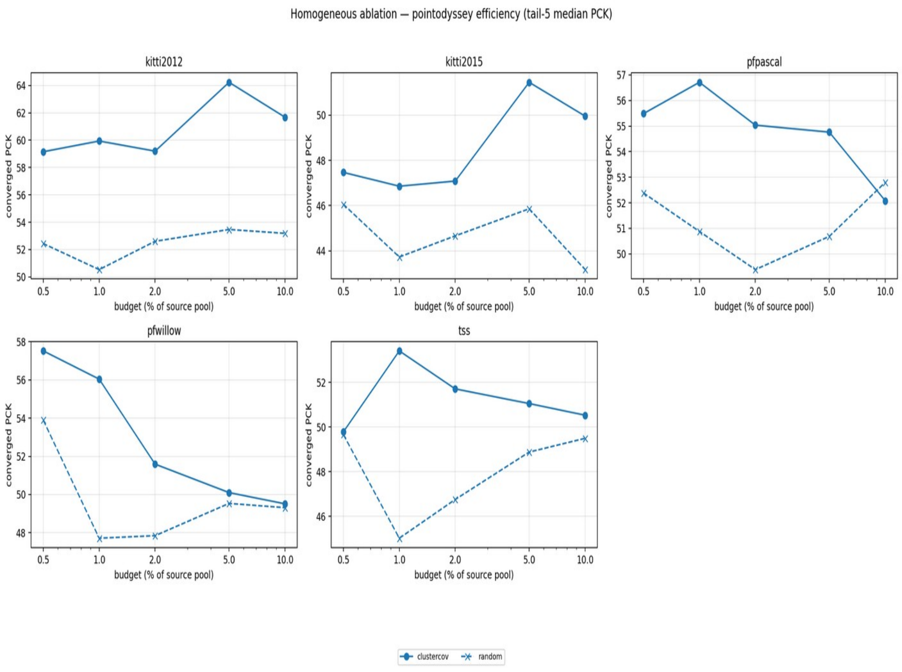


Cluster target motion modes, score new coverage, then grow support across modes greedily.

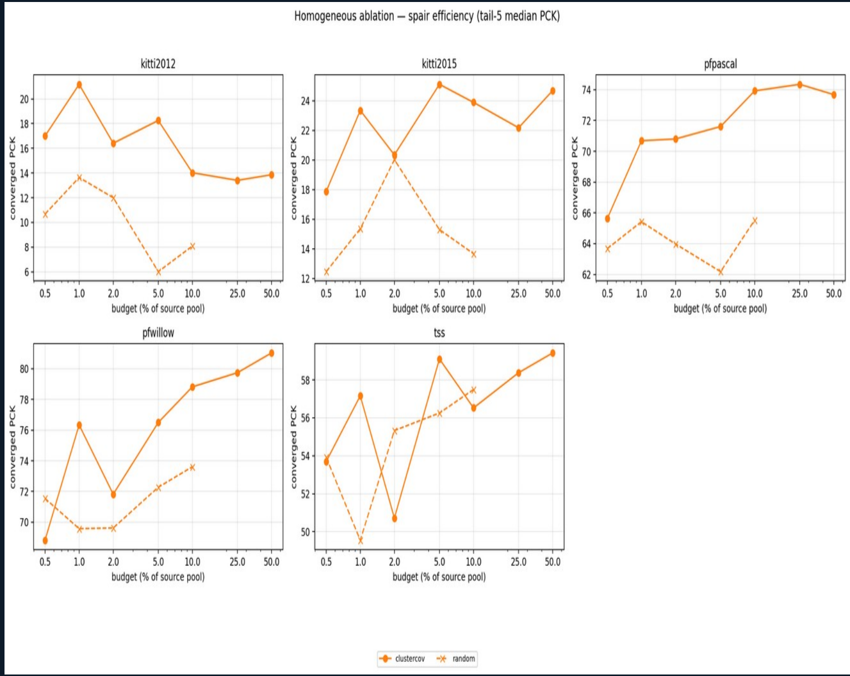
Source Ablations: ClusterCov vs. Random Within Each Source

ClusterCov outperforms random at matched budgets even within a single source — motion-aware selection is sample-efficient · PCK @ α=10%

PointOdyssey pool

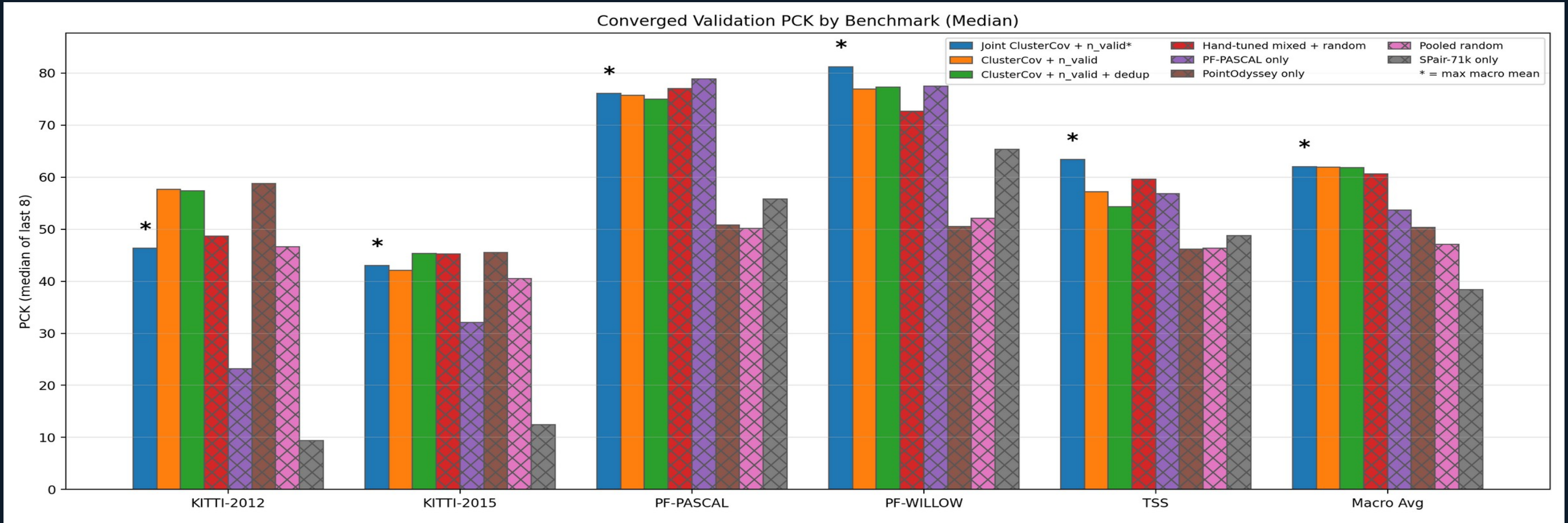


SPair pool



Counterintuitive: BFV coverage beats random on SPair semantic benchmarks — motion geometry encodes cross-domain compatibility even for keypoint data.

Preliminary Results: Converged PCK @ α=10% Across All Benchmarks



What the results show:

- Joint ClusterCov + n_valid leads on TSS and is competitive across all 5 benchmarks
- Single-source baselines (PF-PASCAL-only, PointOdyssey-only) confirm strong Pareto tradeoff
- Hand-tuned mixed baseline competitive on semantic — but requires manual source proportions
- Pooled random substantially below all ClusterCov variants — selection matters

Multi-Target Selector: Automatic Source Routing

Deduplicated union of per-target ClusterCov selections. Routing is learned from motion descriptors alone — no class labels, no appearance features, no source-identity flags.

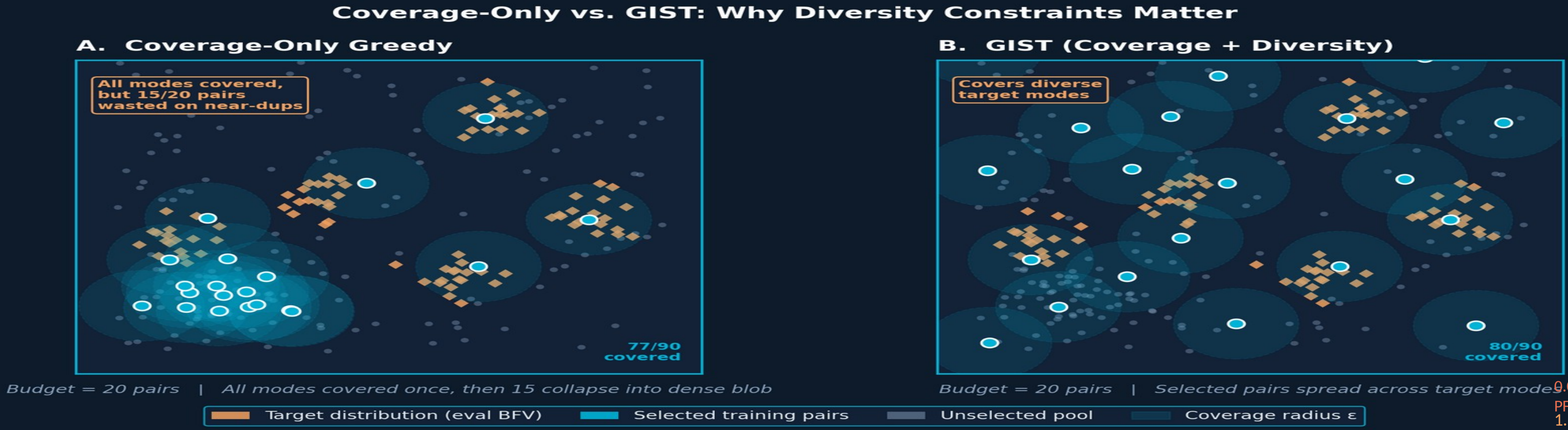
Target	PointOdyssey	SPair-71k	PF-PASCAL
Candidate pool	98.7%	1.2%	0.07%
Dedup union (22,495)	8,386 (37.3%)	13,129 (58.4%)	980 (4.4%)
KITTI-2012 (4,499)	**2,546 (56.6%)**	1,598 (35.5%)	355 (7.9%)
KITTI-2015 (4,499)	**2,376 (52.8%)**	1,924 (42.8%)	199 (4.4%)
PF-PASCAL (4,499)	1,867 (41.5%)	**2,413 (53.6%)**	219 (4.9%)
PF-WILLOW (4,499)	911 (20.2%)	**3,527 (78.4%)**	61 (1.4%)
TSS (4,499)	686 (15.2%)	**3,667 (81.5%)**	146 (3.2%)

Comparison: per-target routing is against a highly imbalanced candidate pool, but the selected union becomes majority semantic.

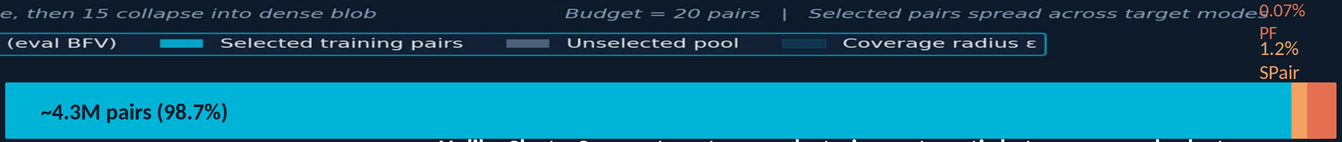
🔑 **Despite a 98.7% PointOdyssey candidate pool, the deduplicated selector shifts to 58.4% SPair. Semantic targets pull semantic sources; geometric targets pull synthetic video — emergent from motion geometry alone.**

GIST: Formal Coverage + Diversity Selection for Correspondence Training Sets

ClusterCov proved coverage-based selection helps — but required k-means clustering and heuristic normalization. GIST replaces both with a formal objective: directed BFV ε-coverage + source-partitioned diversity, with a (½−η) approximation guarantee that extends cleanly to heterogeneous pools.



Heterogeneous pool composition:



Unlike ClusterCov: no target-space clustering, automatic heterogeneous budget allocation, (½−η) guarantee. Same BFV signal that diagnoses transfer now drives principled selection.

3 Correspondence-Specific Adaptations:

- Coverage objective F(S) = BFV E ⊆ε T coverage: fraction of target motion space covered by S within radius ε. Monotone submodular → greedy yields (½−η) guarantee.
- Source-partitioned diversity: penalizes within-source redundancy independently — prevents PointOdyssey (98.7% of pool) from monopolizing budget; lets minority sources (SPair, PF-PASCAL) contribute at appropriate scale.
- Supervision-density-derived ε: scales ε with local correspondence density in target space — dense targets (KITTI flow) get small ε; sparse

Validation Plan Across Chapters

Each chapter makes a concrete, testable claim with chapter-specific metrics and success criteria.

Ch. 1

Cross-Modal Aerial Learning

Validate MAVOC/MAVIC classification on held-out geography, MAVIC-T translation with L1 + LPIPS + FID, and MINNIMAN via map-prediction ablations.

Success: improve on strong baselines and show modality contributions are measurable.

Published / completed

Ch. 2

Directed Coverage Metrics

Validate under held-out target, held-out train set, and joint held-out protocols; compare against symmetric baselines and test controlled SDF motion interventions.

Success: ranking accuracy stays meaningfully above 0.50 and directional metrics beat symmetric ones.

Under review

Ch. 3

Coverage-Based Data Curation

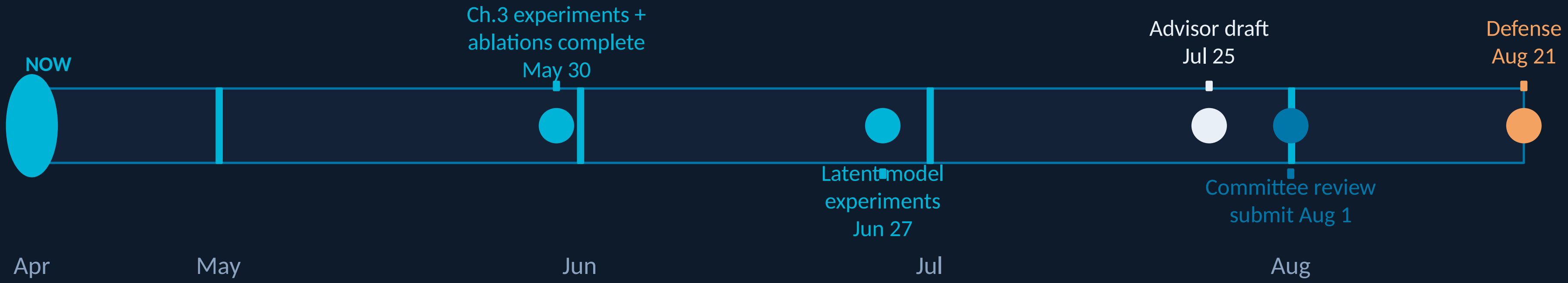
Validate pooled selection, homogeneous source ablations, joint-vs-merged allocation, source routing, GIST ablations, and lambda transfer/sensitivity.

Success: joint or normalized selector gives the strongest macro transfer profile at matched budget.

In progress

Dissertation-level validation claim: robust correspondence improves when training data is explicitly designed, diagnosed, and curated rather than treated as fixed background infrastructure.

Timeline & Status



Chapter 1	MAVOC/MAVIC series + MINNIMAN	**Published (CVPRW 2022–2024)**
Chapter 2	Directed coverage metrics	**Under review (ECCV 2026)**
Chapter 3	Coverage-based data curation	In progress (Defense Aug 21)
	— Source ablations (PointOdyssey, SPair, PF-PASCAL)	In progress (May 30)
	— GIST full framework + λ sweep	**Committee review (Aug 1)**
	— Masked correspondence autoencoder	Planned (stretch)

Dissertation Contributions at a Glance

Ch. 1	• MAVOC/MAVIC benchmark series + MAGIC aligned dataset • SAR ↔ EO translation across 4 modality directions • MINNIMAN: cross-modal magnetic map inference
Ch. 2	• BFV: resolution-agnostic 4D motion descriptor • Directed coverage ($E \subseteq_\varepsilon T$, $T \subseteq_\varepsilon E$): separates failure modes symmetric metrics conflate • SDF-Fractal3D: non-photorealistic pipeline outperforming photorealistic baselines • Transferability estimator: 0.70 pairwise ranking accuracy, 3 held-out protocols
Ch. 3	• Coverage + diversity curation: GIST with BFV submodular utility + ($\frac{1}{2}-\eta$) guarantee • Source-partitioned diversity: handles heterogeneous pool without source labels • Supervision-density-derived ε: removes per-target manual radius tuning • Empirical: motion-only selector recovers expert source routing without labels • [Stretch] Masked correspondence autoencoder for observation-invariant latents

Unifying theme: motion structure is an underused, actionable signal for diagnosing and reducing domain shift in dense visual correspondence.

Anticipated Questions

Q: How sensitive is the method to the normalization exponent β?

A: β derives from target supervision density — dense targets (KITTI) → low β, sparse targets (PF-PASCAL) → high β. Principled, not hand-tuned. Sensitivity ablation planned.

Q: Does λ generalize across target benchmarks?

A: Open question. We plan calibration/deployment split experiment. If yes → offline curation. If no → report as Pareto frontier indexed by λ. Either way is a valid result.

Q: Why not use the learned MAE latent directly for selection instead of BFV?

A: MAE latent clusters by dataset identity (appearance), not motion structure — the very problem we're trying to solve. BFV is the motion-aware alternative; MAE is a downstream representation goal.

Q: Is ClusterCov contribution sufficient if GIST ablations don't show big gains?

A: ClusterCov itself already shows the core claim: subset selection substantially affects downstream transfer. GIST adds formal guarantees and handles heterogeneous pools — important even if marginal empirical gains are modest.

Core framing if pushed: the dissertation's claim is Pareto dominance across benchmarks, not beating every single-target oracle.

Thank You

Bridging Domain Gaps in Visual Correspondence:
From Multi-Modal Fusion to Data-Curated Motion Latents

Ch. 1 ✓ Published

Ch. 2 ✓ Under Review (ECCV)

Ch. 3 In Progress

Defense: August 21, 2026

Questions?

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